Democracy for DAOs

An Empirical Study of Decentralized Governance and Dynamics

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Introduction to DAOs

- Decentralized Autonomous Organizations
- Shift power from centralized authorities to distributed communities.
- Enable collective decision-making through smart contracts and token voting.

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Challenges in DAO Governance

- Voter fatigue and disengagement.
- High operational costs (gas fees)
- Slow decision-making and proposal delays.

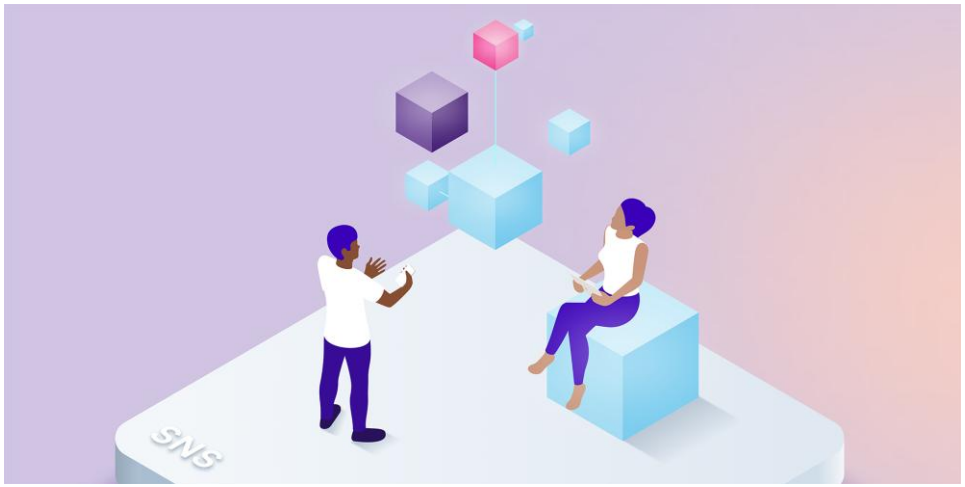


=> Little scalability and effectiveness.

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SNS DAO Framework on ICP

- Liquid democracy: Hybrid of direct and delegated voting.
- Reverse gas model removes tx fees for voters.
- Voting rewards incentivize participation





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Most Active DAOs in the last 30 Days

Name	Proposals	Successful Proposals	Proposals in last 30 days	Successful Proposals in last 30 days
1 WaterNeuron	3391	3182	318	284
2 Network Nervous S...	13443	12607	311	281
3 BOOM DAO	675	604	165	135
4 OpenChat	1902	1733	64	55
5 Dragginz	374	312	58	57
6 CLever	1106	582	46	36
7 AladdinDAO	1161	137	45	0
8 Concentrator	601	280	34	0
9 ICPSwap	680	629		25
10 PokedBots	169	130		18

DefiLlama

DAOs with the Most Proposals

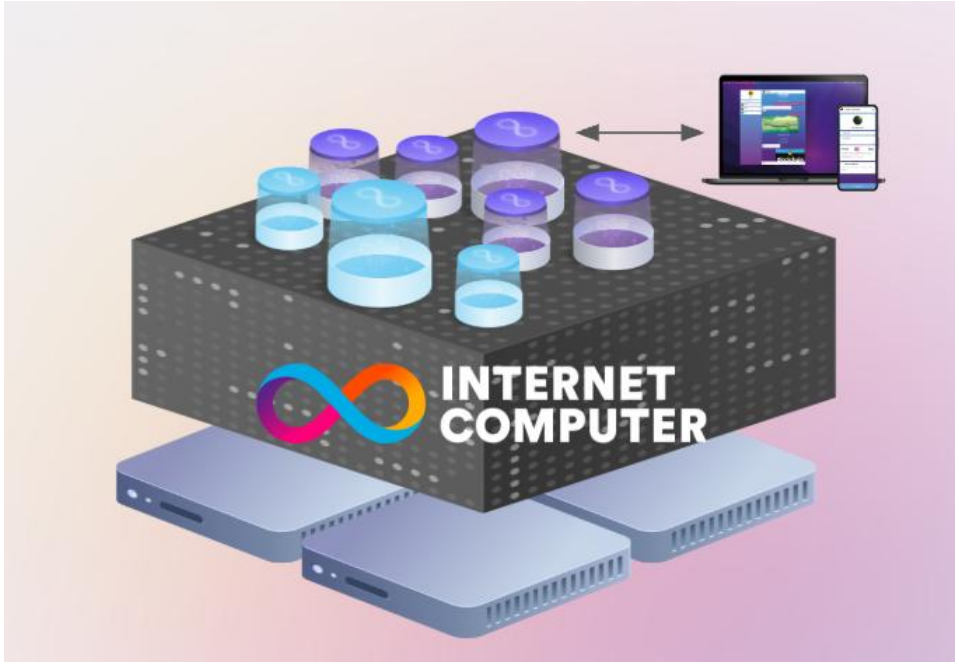
Name	Proposals	Successful Proposals	Proposals in last 30 days	Successful Proposals in last 30 days
1 Network Nervous S...	13443	12607	311	281
2 WaterNeuron	3391	3182	318	284
3 OpenChat	1902	1733	64	55
4 ICFC	1151	1117	11	9
5 Balancer	982	960	20	17
6 Aave	812	776	0	0
7 ICPSwap	680	629	30	25
8 BOOM DAO	675	604	165	135
9 Index Coop	1444	601		0
10 CLever	1106	582		36





Internet Computer Protocol (ICP)

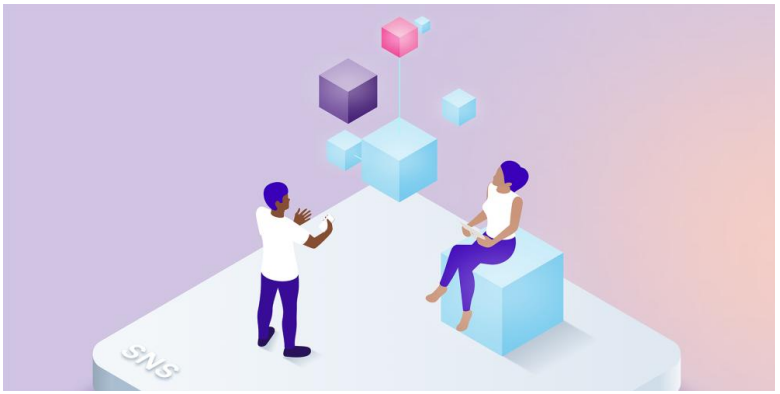
- Decentralized compute platform
- Fast, scalable smart contracts called canisters
- Reverse gas model: paid by developers, not user



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Service Nervous System (SNS)

- Application-specific DAOs running on ICP
- Customizable per DAO.
- Voting power based on locked tokens and lock duration.
- Topic-based delegation and token rewards.



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Research Questions

- How do SNS DAOs perform on participation, approval, and decision speed?
- What are the temporal trends in SNS governance activity?
- Which design features influence these governance dynamics?
- How do SNS DAOs compare with Ethereum DAOs on governance efficiency?

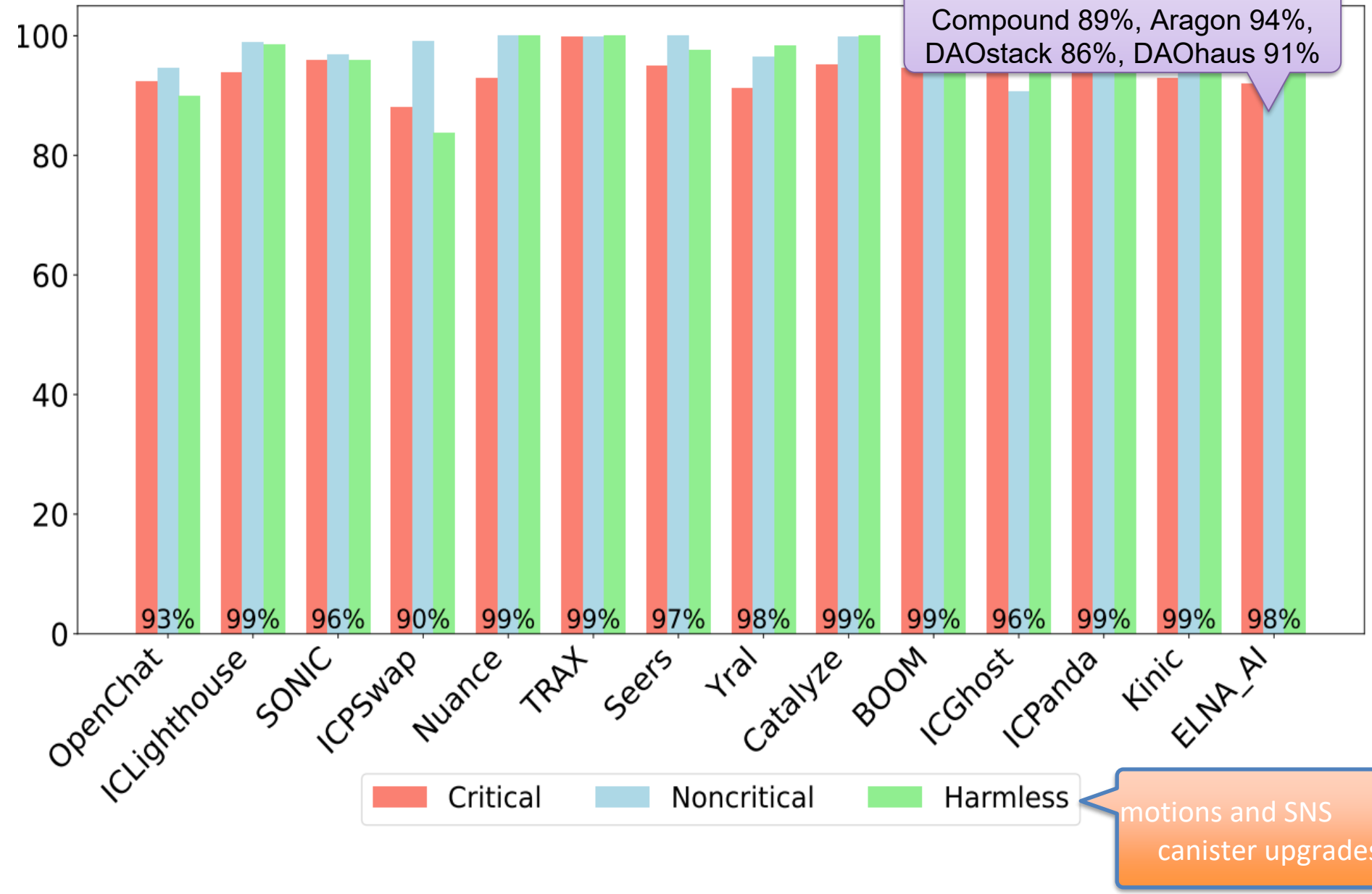
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Data Set

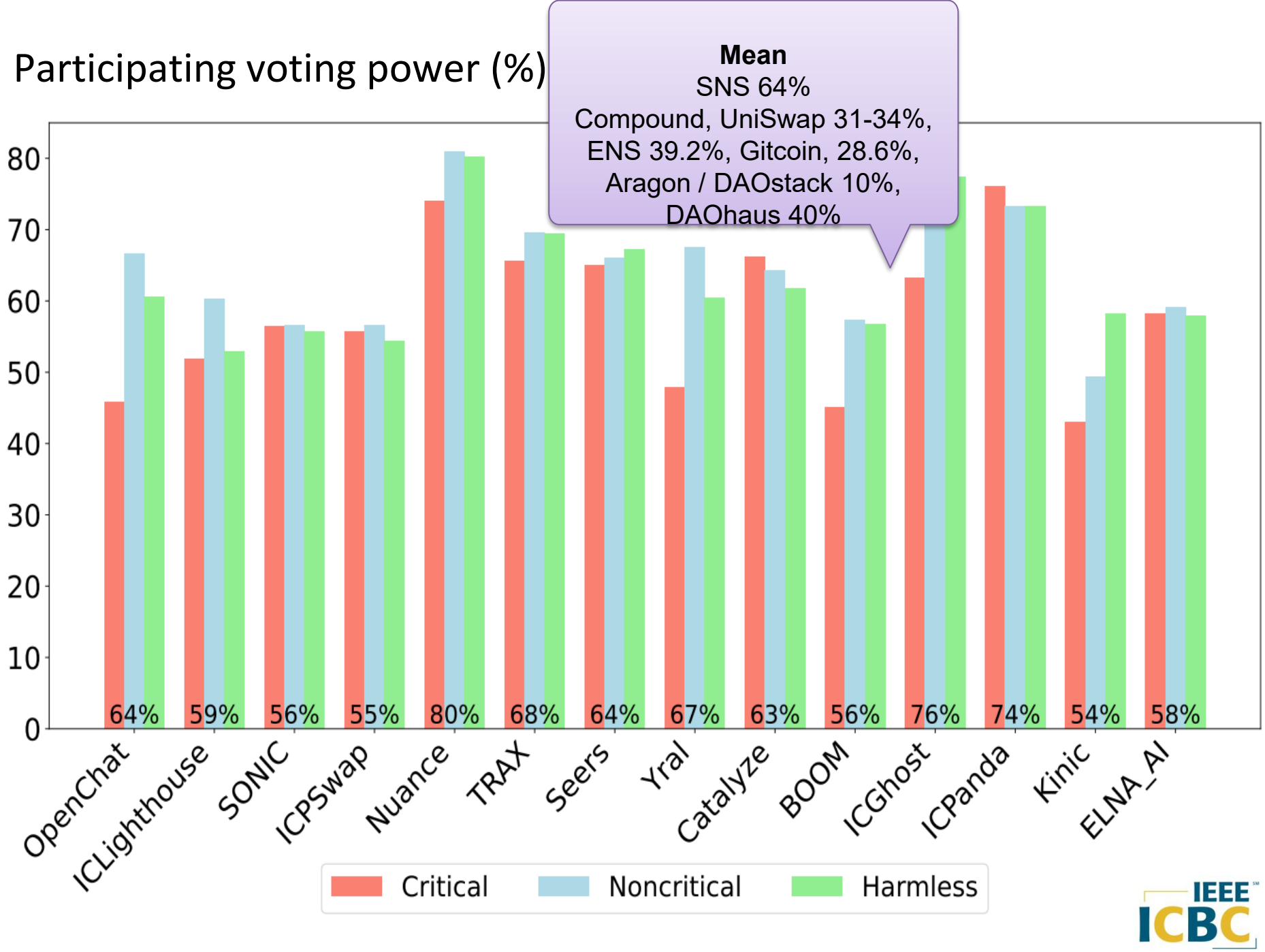
Name	Age months	Category	Treasury USD	Neurons
OpenChat	20	Chat	$ 26,453,292	29,497
ICLighthouse	7	DeFi	$ 41,721,648	7,108
ICPSwap	6	DeFi	$ 11,748,411	10,220
SONIC	12	DeFi	$ 7,236,326	5,132
Nuance	12	Publishing	$ 2,071,079	2,435
TRAX	10	Publishing	$ 6,509,400	4,433
Yral	16	Social Media	$ 7,540,204	16,452
Seers	13	Social Media	$ 5,371,027	2,315
Catalyze	14	Social Media	$ 5,027,497	4,250
BOOM	14	Gaming	$ 2,490,307	5,411
ICGhost	15	Meme Coin	$ 17	2,614
ICPanda	6	Meme Coin	$ 40,525	1,531
Kinic	16	AI	$ 9,025,317	4,918
ELNA	7	AI	$ 6,134,524	5,089

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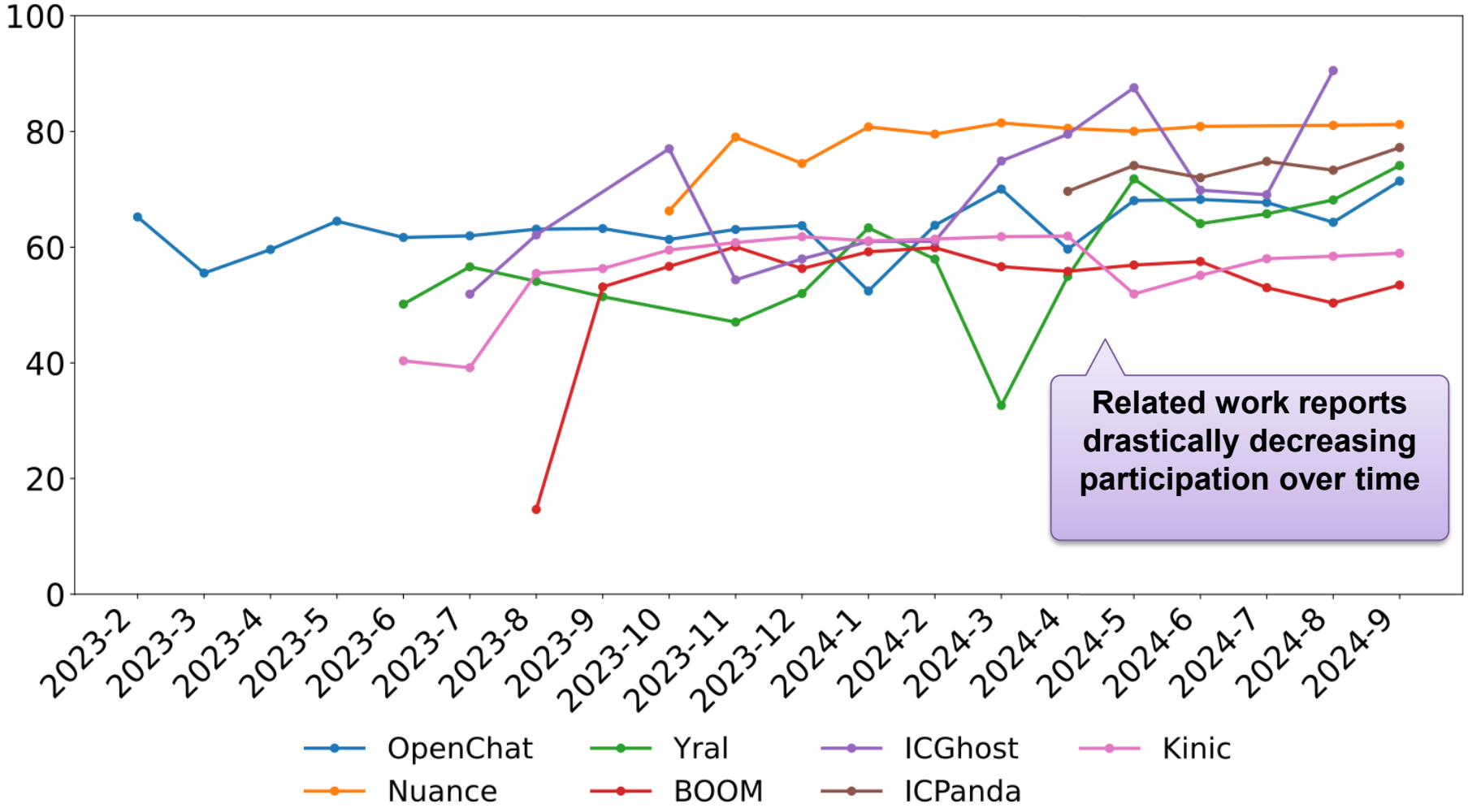
Voting power in favour/total power exercised



Participating voting power (%)

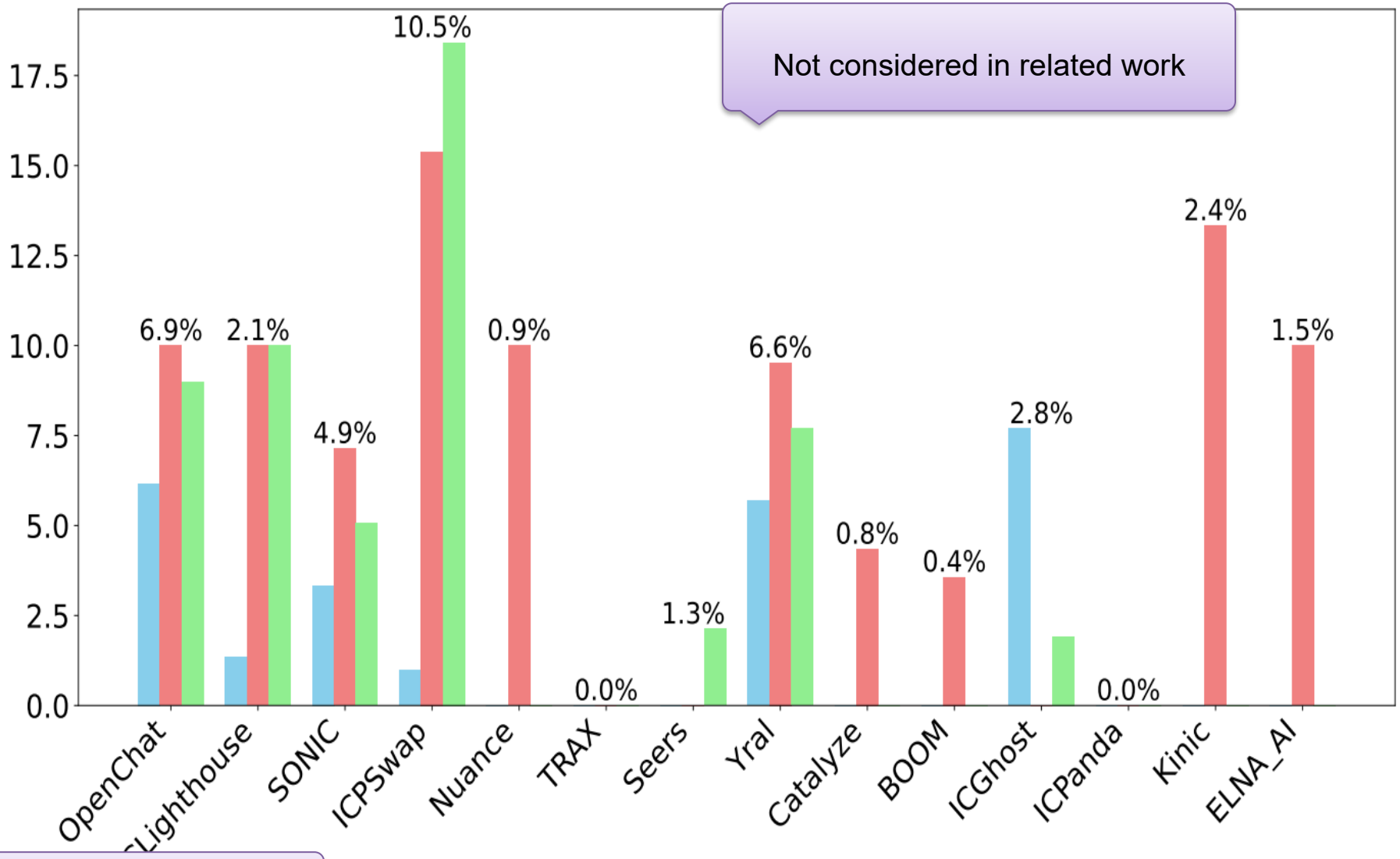


Participating voting power (%)



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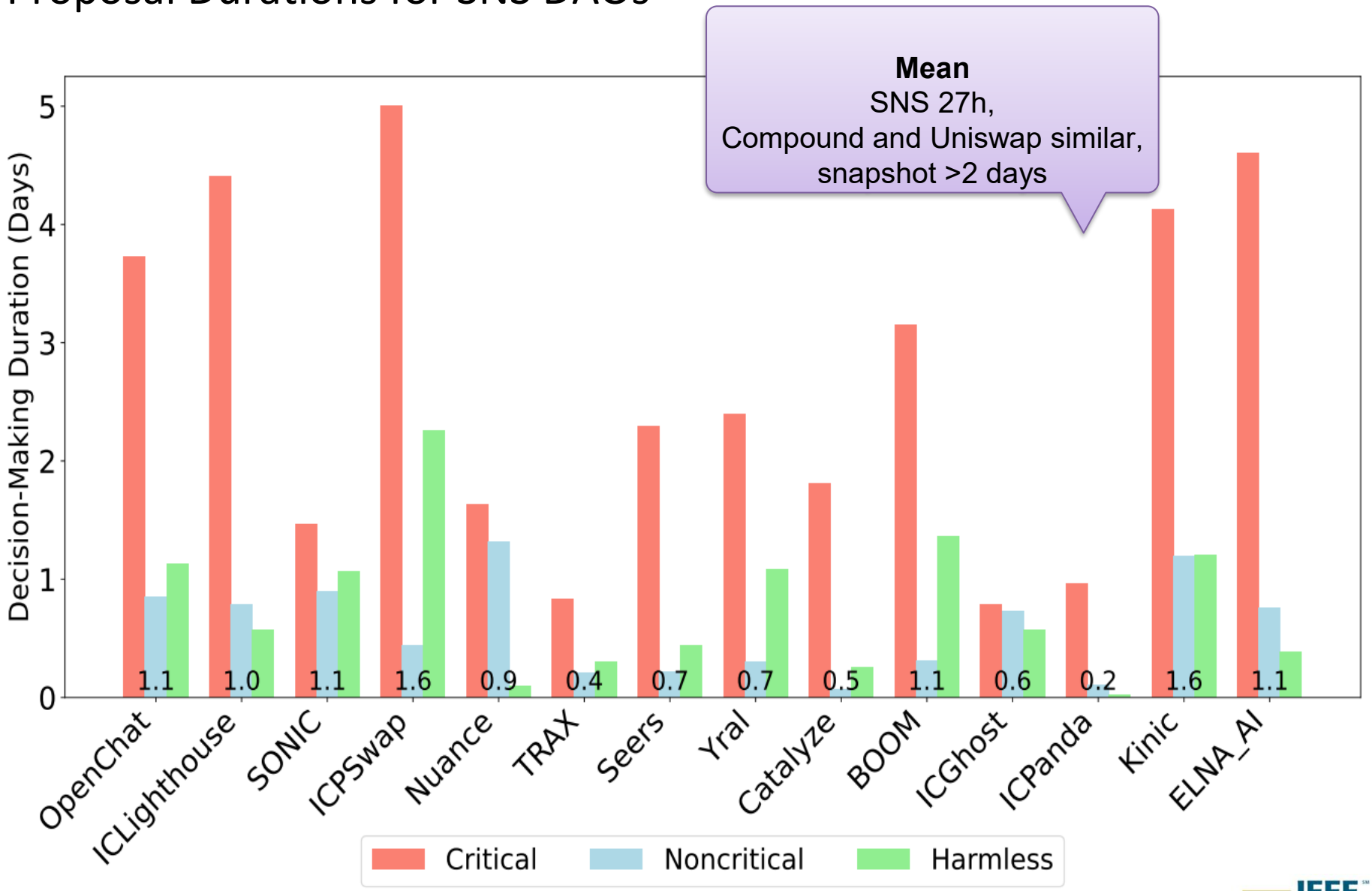
Number of rejected proposals (%)



Only treasury and motion proposals were ever rejected in critical and harmless topics

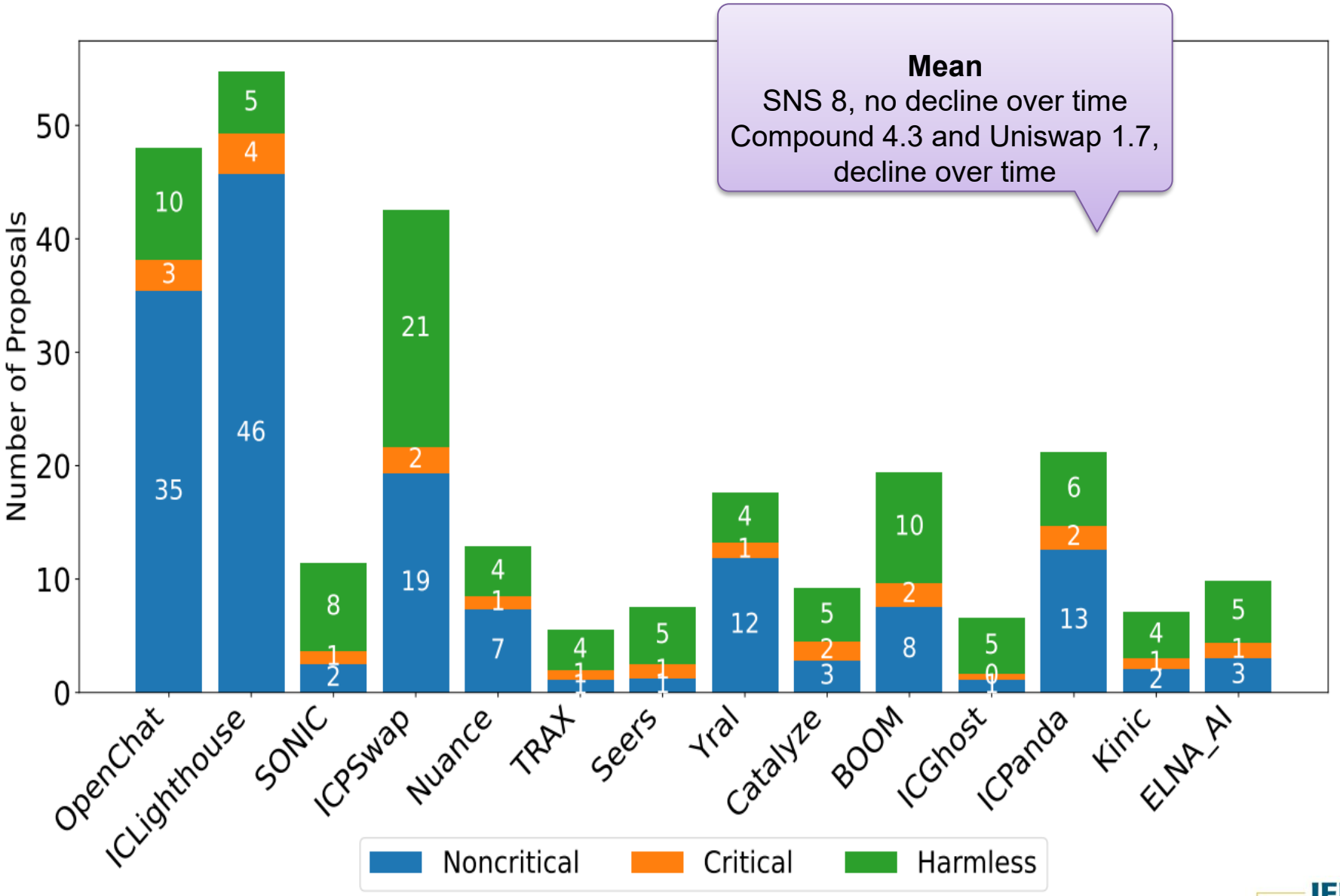
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Proposal Durations for SNS DAOs



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Monthly Frequency of Proposals



Summary

- 14 SNS DAOs
- 3000 proposals
- Metrics
  - participation rates
  - voting power in favour/against
  - frequency of proposals
  - decision duration

Mean cost per proposal
-SNS <11 USD ,
-Gitcoin, Uniswap, Compound
2500-20,000 USD

In comparison to DAOs on other chains:
higher activity, faster decisions, lower costs,
sustained or increasing participation

Questions & Answers

- Thank you for your attention!
- I welcome your questions and discussions.





Democracy for DAOs: An Empirical Study of Decentralized Governance and Dynamics

Case Study Internet Computer SNS Ecosystem

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